

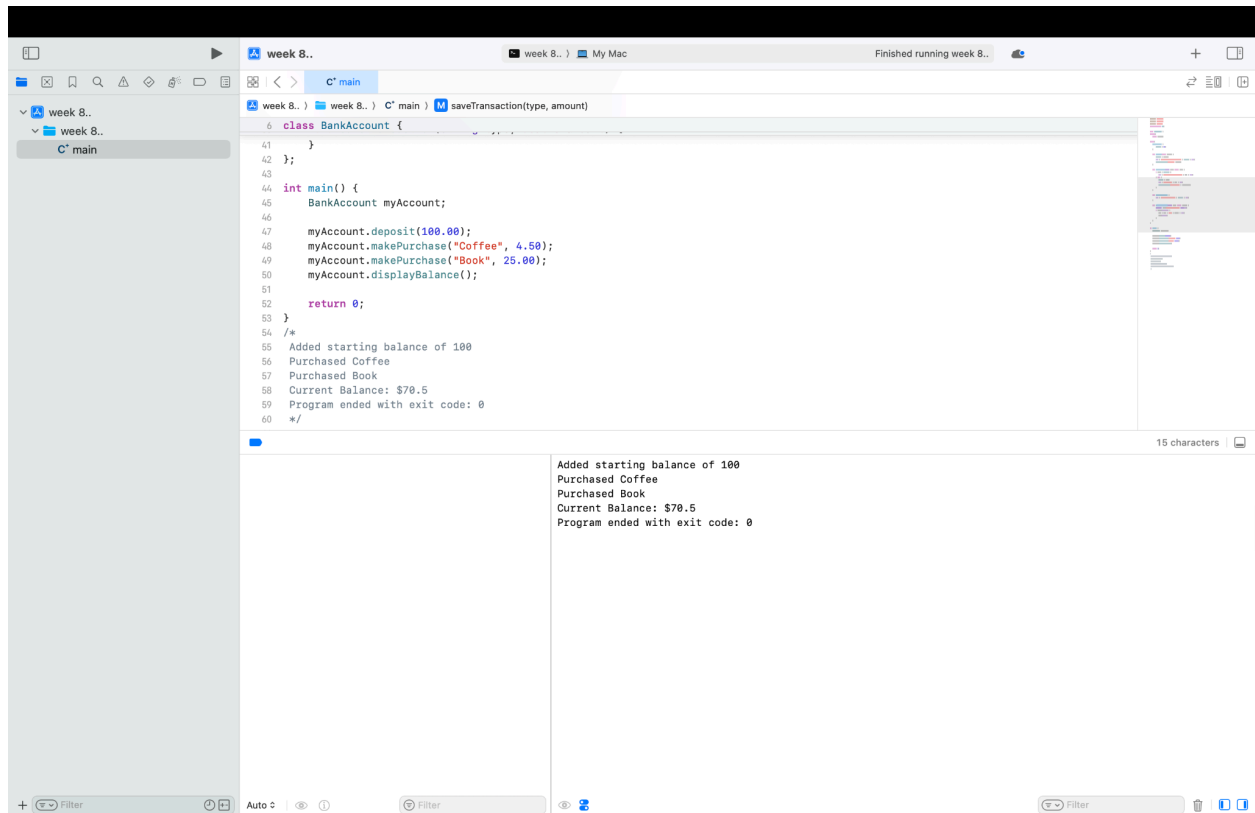
Bank Account Simulation

BankAccount C++ Code

```
1 #include <iostream>
2 #include <fstream>
3 #include <string>
4 using namespace std;
5
6 class BankAccount {
7 private:
8     double balance;
9
10 public:
11     BankAccount() {
12         balance = 0.0;
13     }
14
15     void deposit(double amount) {
16         balance += amount;
17         cout << "Added starting balance of " << balance << endl;
18         saveTransaction("Deposit", amount);
19     }
20
21     void makePurchase(string item, double cost) {
22         if (cost > balance) {
23             cout << "Insufficient funds for " << item << endl;
24         } else {
25             balance -= cost;
26             cout << "Purchased " << item << endl;
27             saveTransaction("Purchase - " + item, cost);
28         }
29     }
30
31     void displayBalance() {
32         cout << "Current Balance: $" << balance << endl;
33     }
34
35     void saveTransaction(string type, double amount) {
36         ofstream file("transactions.txt", ios::app);
37         if (file.is_open()) {
38             file << type << ": $" << amount << endl;
39             file.close();
40         }
41     }
42 };
43
44 int main() {
45     BankAccount myAccount;
46
47     myAccount.deposit(100.00);
```

```
6 class BankAccount {
15 void deposit(double amount) {
16     balance += amount;
17     cout << "Added starting balance of " << balance << endl;
18     saveTransaction("Deposit", amount);
19 }
20
21 void makePurchase(string item, double cost) {
22     if (cost > balance) {
23         cout << "Insufficient funds for " << item << endl;
24     } else {
25         balance -= cost;
26         cout << "Purchased " << item << endl;
27         saveTransaction("Purchase - " + item, cost);
28     }
29 }
30
31 void displayBalance() {
32     cout << "Current Balance: $" << balance << endl;
33 }
34
35 void saveTransaction(string type, double amount) {
36     ofstream file("transactions.txt", ios::app);
37     if (file.is_open()) {
38         file << type << ": $" << amount << endl;
39         file.close();
40     }
41 }
42 };
43
44 int main() {
45     BankAccount myAccount;
46
47     myAccount.deposit(100.00);
48     myAccount.makePurchase("Coffee", 4.50);
49     myAccount.makePurchase("Book", 25.00);
50     myAccount.displayBalance();
51
52     return 0;
53 }
54
55 /*
56 Added starting balance of 100
57 Purchased Coffee
58 Purchased Book
59 Current Balance: $70.5
60 Program ended with exit code: 0
61 */
```

Program Output



```
1 // week 8...
2 #include <iostream>
3 using namespace std;
4
5 // BankAccount class
6 class BankAccount {
7     private:
8         double balance;
9     public:
10        BankAccount() { balance = 0; }
11        BankAccount(double b) { balance = b; }
12        void deposit(double amount) { balance += amount; }
13        void makePurchase(string item, double amount) { balance -= amount; }
14        void displayBalance() { cout << "Current Balance: " << balance << endl; }
15};
16
17 int main() {
18     BankAccount myAccount;
19     myAccount.deposit(100.00);
20     myAccount.makePurchase("Coffee", 4.50);
21     myAccount.makePurchase("Book", 25.00);
22     myAccount.displayBalance();
23     return 0;
24}
25
26 /*
27 Added starting balance of 100
28 Purchased Coffee
29 Purchased Book
30 Current Balance: $70.5
31 Program ended with exit code: 0
32 */
33
```

Added starting balance of 100
Purchased Coffee
Purchased Book
Current Balance: \$70.5
Program ended with exit code: 0

Transactions.txt File

